

Opg 1

a) $f(x) = \frac{\pi}{4} - \frac{x}{2}, \quad 0 < x < \pi \Rightarrow L = \pi$

$$a_0 = \frac{1}{L} \int_0^L f(x) dx$$

$$= \frac{1}{\pi} \int_0^{\pi} \left(\frac{\pi}{4} - \frac{x}{2} \right) dx$$

$$= \frac{1}{\pi} \left[\frac{\pi x}{4} - \frac{x^2}{4} \right]_0^{\pi}$$

$$= \frac{1}{\pi} \left(\frac{\pi^2}{4} - \frac{\pi^2}{4} \right) = 0$$

$$a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \left(\frac{\pi}{4} - \frac{x}{2} \right) \cos(nx) dx$$

$$= \frac{1}{2} \int_0^{\pi} \cos(nx) dx - \frac{1}{\pi} \int_0^{\pi} x \cos(nx) dx$$

$$= \frac{1}{2} \left[\frac{\sin(nx)}{n} \right]_0^{\pi} - \frac{1}{\pi} \left[\frac{x \sin(nx)}{n} - \frac{1}{n} \int \sin(nx) dx \right]_0^{\pi}$$

$$= -\frac{1}{\pi} \left[0 + \frac{\cos(nx)}{n^2} \right]_0^{\pi}$$

$$= -\frac{1}{\pi} \left(\frac{\cos(\pi n)}{n^2} - 1 \right)$$

$$= \frac{1 - (-1)^n}{\pi n^2}$$

$$= \begin{cases} \frac{2}{\pi n^2}, & \text{odd} \\ 0, & \text{even} \end{cases}$$

$$f(x) \sim a_0 + \sum_{n=1}^{\infty} a_n \cos(nx)$$

$$= \sum_{n=0}^{\infty} \frac{2}{\pi (2n+1)^2} \cos((2n+1)x)$$

$$= \frac{2}{\pi} \sum_{n=0}^{\infty} \frac{\cos((2n+1)x)}{(2n+1)^2}, \quad 0 < x < \pi$$

b) $g(x) = \begin{cases} \frac{\pi}{4} - \frac{x}{2}, & 0 \leq x \leq \pi \\ \frac{\pi}{4} + \frac{x}{2}, & -\pi \leq x \leq 0 \end{cases}$

$$x=0: \quad \frac{\pi}{4} = \sum_{n=0}^{\infty} \frac{2}{\pi (2n+1)^2} \Rightarrow \sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$$

Opg 2

$$\int_{-\infty}^{\infty} f(x-t) e^{-itx} dt = e^{ix^2}$$

ser at os e

. Tar Fouriertrans. av begge sider og får

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x) g(x-t) dt \Rightarrow F(f * g) = \sqrt{2\pi} F(f) F(g)$$

$$F\left(\int_{-\infty}^{\infty} f(x-t) e^{-it} dt\right) = F(e^{-\frac{x^2}{2}})$$

$$\sqrt{2\pi} F(f) F(e^{-it}) = F(e^{-\frac{x^2}{2}})$$

$$F(f) = \frac{1}{\sqrt{2\pi}} \frac{1}{F(e^{-it})} F(e^{-\frac{x^2}{2}})$$

$$= \frac{1}{\sqrt{2\pi}} \frac{\sqrt{\pi}}{\sqrt{2}(\omega^2+1)} e^{-\frac{\omega^2}{2}}$$

$$F(f) = \frac{1}{2} \omega^2 e^{-\frac{\omega^2}{2}} + \frac{1}{2} e^{-\frac{\omega^2}{2}}$$

$$F' = (\omega F(f)) \Rightarrow F^{-1}(f) = \frac{1}{\omega^2} F^{-1}\left(\frac{\partial f}{\partial \omega}\right)$$

$$F^{-1}\left(\frac{1}{2} e^{-\frac{x^2}{2}}\right) = \frac{1}{2} e^{-\frac{x^2}{2}}$$

$$F(f) = \frac{1}{2} \omega^2 e^{-\frac{\omega^2}{2}} \Rightarrow F^{-1}(0)$$

Oppg 3

$$y'' - 3y' + 2y = 0, \quad y(0) = 1 \text{ og } y'(0) = 0$$

$$(s^2 F(s) + s y(0) + y'(0)) - 3(s F(s) + y'(0)) + 2 F(s) = 0$$

$$s^2 F(s) + s - 3s F(s) - 3 + 2 F(s) = 0$$

$$s^2 F(s) - 3s F(s) + 2 F(s) = 3 - s$$

$$F(s)(s^2 - 3s + 2) = 3 - s$$

$$F(s) = \frac{3-s}{s^2-3s+2} = \frac{3-s}{(s-1)(s-2)} = \frac{A}{s-1} + \frac{B}{s-2} \Rightarrow 3-s = A(s-2) + B(s-1) \Rightarrow 3-s = -2A-B+s(A+B)$$

$A = -2, B = 1$

$$F(s) = \frac{1}{s-2} - \frac{2}{s-1}$$

$$f(t) = e^{2t} - 2e^t \quad 2e^t - e^{2t}$$

b) Vi setter $x(t) = y'(t)$ s.a. vi får

$$x'(t) = 3x(t) - 2y(t)$$

$$y'(t) = x(t)$$

Vi setter $h=0,1$ og def x_n og y_n som tilnærminger til $x(nh)$ og $y(nh)$. Da får vi

$$x_{n+1} = x_n + h x'_n = x_n + h(3x_n - 2y_n)$$

$$y_{n+1} = y_n + h x_n$$

$$x_0 = 0$$

$$y_0 = y(0) = 1$$

$$x_1 = x_0'' + h(3x_0'' - 2y_0) = -2y_0 h = -0,2$$

$$y_1 = y_0 + h x_0'' = 0$$

Oppg 4

$$e^{\frac{x}{3}} - x = 0, \quad x \in (0, 3), \quad x_0 = 1$$

$$x = e^{\frac{x}{3}} = g(x)$$

$$x_{k+1} = g(x_k) = e^{\frac{x_k}{3}}$$

$$x_1 = g(x_0) = e^{\frac{1}{3}} = 1,395612$$

$$x_2 = e^{\frac{x_1}{3}} = 1,592339$$

$$x_3 = e^{\frac{x_2}{3}} = \underline{\underline{1,700257}}$$

Oppg 6

$$u_t = u_{xx}, \quad 0 \leq x \leq 3, \quad t \geq 0$$

a) $u(x, t) = F(x)G(t)$

$$F(x)G'(t) = F''(x)G(t)$$

$$\frac{F''(x)}{F(x)} = \frac{G'(t)}{G(t)} = \kappa$$

$$F''(x) - \kappa F(x) = 0 \Rightarrow F''(x) + p^2 F(x) = 0, \quad \kappa = -p^2 \text{ to make sure } \kappa < 0$$

$$F(x) = A \cos(px) + B \sin(px), \quad p = \frac{n\pi}{L}$$

$$F(0) = A = 0$$

$$F(3) = B \sin\left(\frac{n\pi \cdot 3}{L}\right) = 0, \quad B \neq 0$$

$$F_n(x) = B \sin\left(\frac{n\pi x}{L}\right)$$

$$G'(t) + p^2 G(t) = 0 \Rightarrow G(t) = G_0 e^{-p^2 t}$$

$$u(x, t) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) e^{-\frac{n^2 \pi^2}{L^2} t}$$

b) $u(x, 0) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{L}\right) = 2 \sin\left(\frac{\pi x}{3}\right) \Rightarrow B_1 = 2$

$$\underline{u(x, t) = 2 \sin\left(\frac{\pi x}{3}\right) e^{-\frac{\pi^2}{9} t}}$$

